

Advances in surface-enhanced Raman spectroscopy-based sensors for detection of various biomarkers

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ABSTRACT

Surface enhanced Raman spectroscopy (SERS) allows the ultrasensitive detection of analytes present in traces or even single molecule levels by the generation of electromagnetic fields. It is a powerful vibrational spectroscopic method that is capable to detect traces of chemical and biological analytes. SERS technique is involved in the extremely sophisticated studies of molecules with high specificity and sensitivity. In the vicinity of nanomaterials decorated surfaces, SERS can monitor extremely low concentrations of analytes in a non-destructive manner with narrow line widths. This review article is focused on some recently developed SERS-based sensors for distinct types of analytes like disease-related biomarkers, organic and inorganic molecules, various toxins, dyes, pesticides, bacteria as well as single molecules. This study aims to enlighten the arising sensing approaches based on the SERS technique. Apart from this, some basics of the SERS technique like their mechanism, detection strategy, and involvement of some specific nanomaterials are also highlighted herein. Finally, the study concluded with some discussion of applications of SERS in various fields like food and environmental analysis.

1. Introduction

Surface Enhanced Raman Spectroscopy or SERS is a highly sensitive and powerful technique based upon vibrational spectroscopy that allows the detection of very low concentrations of various types of biological samples and diseases. SERS is extremely capable of identifying chemical species and obtaining their structural information which includes a variety of fields such as material and polymer science, catalysis, electrochemistry, biochemistry, and biosensing (Liang et al., 2021; McNay et al., 2011; Ogundare and van Zyl, 2019; Perumal et al., 2021). The detection is mainly based on the amplification of electromagnetic signals generated through localized surface Plasmon's excitation (Prochazka and Prochazka, 2016; Etchegoin and Le Ru, 2010; Sharma et al., 2012). The technique is supported by nanomaterials and increases the Raman scattering of molecules [Fig. 1]. Martin Fleischmann, A. James McQuillan, and Patrick J. Hendra first observed SERS in 1973, and since then its utilization has grown exponentially. After the first observation of Raman spectra on roughened electrodes made up of silver, the technique has dramatically progressed in sensing, single molecule detection, imaging, and ultrafast science (Fleischmann et al., 1973). Several methods have been implemented to enhance the efficiency of Raman

scattering by the researchers like electronic resonance enhancement and Raman processes stimulation, but the SERS provides the most significant signal amplification of Raman processes. Basically, SERS increases the weak signals significantly from the Raman scattering technique (Littleford et al., 2004; Kumar et al., 2020).

In the SERS process, the electromagnetic enhancement mechanism is the dominant contributor. The excitation of localized surface plasmon resonance (LSPR) amplifies the light which results in the enhancement of electromagnetism (Weihua et al., 2009). This light concentration is present in the gaps or sharp features of nano-structured plasmonic materials such as silver, copper, and gold. The electromagnetic enhancement also depends upon the structure of plasmonic materials for SERS. Another mechanism that may be exploited for signal enhancement is known as chemical enhancement which involves primarily charge transfer by excitation wavelength. The SERS enhancement factor may approach $\sim 10^{10}$ – 10^{11} for highly optimized surfaces that are the product of chemical and electromagnetic enhancement mechanisms (Pilot et al., 2019). For ultrasensitive detection through the SERS technique, some effective factors are (a) Detailed knowledge of analytes must be known i. e., whether they are inorganic ions or molecules. This knowledge is essential to knowing the Raman scattering properties. Other parameters

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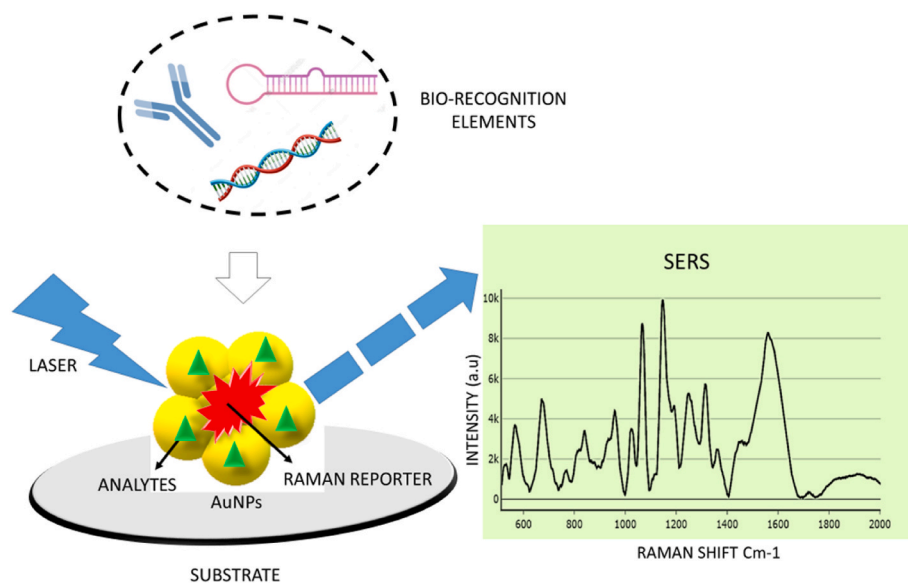


Fig. 1. A schematic representation of the development of SERS-based sensing platform decorated with Raman reporter with nanomaterials for enhancement of signals.

which also affect the sensing of the molecule are molecular structure, molecular weight, electronegativity, photoluminescence, water solubility, and their affinity for nanomaterials. On the other hand, for inorganic ions that are not showing Raman scattering cross-section, labeling with a suitable reporter is a must. (b) Visible, ultraviolet, and near-infrared excitation window of nanoparticles (NPs) and incident wavelength for Raman scattering excitation. (c) The adsorption mechanism of analytes with SERS substrate also plays an important role due to their influence on SERS intensity. (d) Choose different strategies for in vitro and in vivo detection (Yuan et al., 2017). Overall, the SERS technique has been successfully utilized for the ultrasensitive detection of chemicals and molecules in the field (Kneipp et al., 2008; Le Ru and Etchegoin, 2012). Additionally, SERS involved in ultrafast spectroscopies, photocatalytic and photovoltaic reactions (Prakash, 2019; Wang et al., 2020). Mechanistic studies on stimulated SERS are relatively unexplored, and some predict that enhancement factors could exceed 10^{20} . Many of these were limited to one vibrational peak with surface enhancement factors estimated to be well under 10^4 (Sharma et al., 2012; Byram et al., 2019).

2. Importance of nanomaterials in SERS-based detection

Nanomaterials-based SERS detection of biomolecules gained much attention over the past few years. Different dimensional nanomaterials are being utilized as a substrate to enhance the features of SERS to achieve ultrasensitive detection. Variations in properties of nanomaterials such as size, composition, morphology, stability, dielectric properties, and surface modifications play an important role in the generation of enhanced electromagnetic signals (Guo and Dong, 2011; Du et al., 2021). Various noble metal NPs have been studied for their SERS activity such as gold, graphene/graphene oxides, molybdenum disulfide, black phosphorous, and silver (Howes et al., 2014; Huang et al., 2019; Karthick Kannan et al., 2019). Gold is the most exploited nanomaterial in SERS-based detection due to its properties like unique electronic structure, LSPR absorption properties, biocompatibility, and excellent chemical stability. Gold NPs in less than 100 nm range exhibit strong electromagnetic behavior when irradiated by excitation light. It was also investigated that the Raman intensity of analytes also increases by order of magnitude when analytes come in contact with gold NPs. Thus, adding gold nanomaterial can improve substance detection efficiency significantly. The irregular geometry of gold NPs, including

spike-like tips, dimers, and aggregates with small nanogaps, plays a pivotal role in enhancing the specificity and sensitivity of SERS. These features create plasmon coupling and Raman hot spots, leading to significant signal amplification (Barbillon, 2020). Besides gold, silver nanoparticles are also used widely in the SERS effect and exhibit superior enhancement in the Raman Effect (Herrera et al., 2013). Silver NPs showed a strong electromagnetic field under laser irradiation between each particle. Metal NP-dependent SERS detection exhibits high sensitivity due to enhancement in the SERS intensity of molecules. The metal NPs show uneven tips and surfaces that can enhance electromagnetism to amplify the Raman signals of adjacent molecules/substances (Cao et al., 2012). Among various types of nanomaterials, graphene and graphene oxides attract researchers for SERS-based biosensing. Graphene exhibits superior adsorption capacity and excellent physical strength. Other nanomaterials that are employed for SERS detection are molybdenum disulfide, black phosphorous, titanium oxide, and zinc oxide (Zong et al., 2018). An important factor for controlling the NPs aggregation process is their surface chemistry, i.e., ligands that stabilize the NPs may determine the aggregation state, as well as the rate of aggregation in colloidal system physicochemical interactions between the analyte and stabilizing ligand molecules, which are triggered. The surface modification by organic ligands provides stability against aggregation as well as introduces a variety of surface functionalities available for further reactions. Gold NPs were also applied as tools for the detection of extremely small amounts of analytes of wider interest such as pollutants, toxins, and small molecules (Krpetic et al., 2012). Similarly, gold NPs have been applied in the SERS-based detection of enzymes, DNA, and cancer antigens (Chen et al., 2017).

3. Functionalization and immobilization of biomarkers

Functionalizing surfaces in the SERS technique involves the immobilization or adsorption of various biomolecules and chemicals to enhance the specificity and selectivity of analyte detection. Proteins, enzymes, and antibodies can be immobilized onto SERS-active surfaces to create specific recognition sites for target analytes. This immobilization can be achieved through physical adsorption, covalent bonding, or cross-linking techniques. The functionalized surfaces provide a platform for capturing and detecting analytes with high affinity and selectivity. Furthermore, single-stranded DNA, RNA and/or aptamers can be adsorbed onto SERS-active surfaces. The DNA-functionalized surfaces

Table 1

Recently fabricated SERS-based sensor for different types of analytes.

Analytes	Nanomaterial used	Limit of detection	Linear range	References
Rhodamine 6G Crystal violet	Silver	10^{-5} M 10^{-4} M	NR	40
Cardiac troponin I	Citrate-capped gold-silver bimetallic NPs, gold core with silver shell NPs, and rattle-like gold core in silver-gold shell NPs	0.09 ng/mL	0.09–50 ng/mL	41
Exosomes	ZnO nanorods and silver	≈20 exosomes/ μ L	10^4 to 10^8 exosomes/ μ L	42
Uric acid	AuNPs/PAN/L electrospun nanofiber composites	10^{-7} M	10^{-3} – 10^{-4} M	43
Gestational diabetes mellitus	Anisotropic gold nanotriangles	3.0×10^{-16} g/mL	10^{-15} – 10^{-6} g/mL	46
Prostate cancer biomarkers	Raman dyes encoded core-shell SERS nanotags	0.37, 0.43, and 0.26 pg/mL for PSA, CEA, and AFP, respectively	For PSA 1 pg/mL up to 10 μ g/mL For both CEA and AFP- 10 pg/mL to 1 μ g/mL	48
Prostate-specific antigen	Silver	NR	0.1–20 ng/mL for f-PSA and 1–200 ng/mL for t-PSA	49
Prostate-specific antigen	Silver	6 pg/mL	0.25–10 ng/mL	50
Colorectal cancer	Au core-Ag shell nanoparticles	0.82 pg/mL	1 pg/mL - 1 mg/mL	53
Kanamycin	Gold and silver	0.90 pg/mL	10 μ g/mL to 100 ng/mL	62
Rhodamine 6G and crystal violet	Silver	10^{-10} M	1 μ M to 100 pM	64
Mercury	Gold	60 aM	NR	68
Mercury	Silver	0.1 ppb	1–100 ppb	69
Microcystin-LR	Gold	0.002 ng/mL	0.01–200 ng/mL	70
Ochratoxin-A	Silver	2.63 pg/mL	10^3	71
Aflatoxin-B1		4.15 pg/mL	10^{-5} μ g/mL	
Pesticides	Gold	0.79 pg/mL, 1.04 pg/mL, 1.21 pg/mL	10^{-2} to 2×10^3 ng/mL for 2, 4-D And 0.1–1000 ng/mL for Pymetrozine, Thiamethoxam	78
2,4-dichloro phenoxy acetic acid, Pymetrozine, Thiamethoxam				
Pesticide acetamiprid	rGO and Gold	2.13×10^5 μ g/mL	1.0×10^{-4} to 1.0×10^3 μ g/mL	79
Bacteria <i>Y. pestis</i> , <i>F. tularensis</i> , and <i>B. anthracis</i>	Gold	43.4, 45.8, and 357 CFU/mL	0– 10^7 CFU/mL	85
<i>E. coli</i> O157:H7	Gold	NR	10^0 – 10^5 CFU/mL	86
IgG	Ag–Au-assembled silica NPs	0.09 ng/mL	0.6–9.0 ng/mL	87

*NR-Not Reported.

can be designed to hybridize with complementary DNA sequences or specific target analytes, allowing for their detection through SERS. DNA molecules can be adsorbed onto SERS-active surfaces, either as single strands or in the form of double-stranded structures. Similarly, peptide and various chemicals or functional groups that have specific interactions with target analytes, can be used to functionalize SERS-active surfaces. These chemicals may include chelating agents, surface ligands, or reactive functional groups that can form complexes or covalent bonds with the analytes of interest (Nsiah and McDermott, 2023).

4. Recently developed SERS-based detection systems

SERS-based sensors have gained much attention in the last decade for the detection of distinct types of analytes in the field of life science and analytical chemistry. LSPR hot spots enable the enhancement of electromagnetic field which is responsible for SERS enhancement with an enhancement factor of approximately 10^{12} (Rodrigues et al., 2015). Hot spots are formed typically at small gaps (approximately 10 nm) between noble metal NPs (Cai et al., 2021). The SERS method in combination with other assays and NPs, draws increasing interest among the researchers for ultrasensitive detection of analytes (Guo et al., 2020; Li et al., 2010; Zhang et al., 2017; Kim et al., 2021). The SERS method has emerged as a highly sensitive and accurate technique for fingerprint recognition and molecular sample analysis. To detect various types of analytes such as pesticide residues and water pollutants, as well as for safety purposes, a portable Raman spectrometer proves to be an excellent choice. In some cases, SERS sites are created to effectively separate gold nanorods from each other using shells made of a metal-organic framework. This approach aids in achieving exceptional quantification capabilities. However, Ding and their team presented a challenge to this strategy by opting for an analyte enrichment function to compensate for

the absence of SERS sites. By implementing this approach, the capability of analyte detection through SERS-based sensors was significantly enhanced. The reported sensing technique involved processes such as analyte filtration, enrichment, size-selective identification, and remarkably high sensitivity (Fu et al., 2016; Ngo et al., 2013; Ding et al., 2020). In this context, several experiments have been performed in recent years. Some studies are discussed below and summarized in Table 1.

4.1. SERS-based detection of disease-related biomarkers

SERS is a powerful analytical technique that can be used for the detection of disease-related biomarkers. It combines the principles of Raman spectroscopy with the unique properties of nanostructured metal surfaces to enhance the Raman signal of molecules adsorbed on or near the surface. A paper-based SERS chip was fabricated as Raman spectral analyzer by utilization of silver NPs and integrated with the smartphone. At present, this type of portable device is in demand which does not require any skilled person to test samples. On the other hand, these devices are also cost-effective as replace high-cost instruments (ZengMou et al., 2018). Another point-of-care device was developed by combining lateral flow assay strips with SERS for quantitative analysis of biomarkers in a very low concentration range. Three types of NPs have been tested in this study viz. Citrate-capped gold-silver bimetallic NPs, gold core with silver shell NPs, and rattle-like gold core in silver-gold shell NPs. The citrate-capped gold-silver metallic NPs have shown the highest SERS activity among all three probes whereas gold core with silver shell NPs was used in a SERS-based lateral flow assay strip as optical as well as SERS probes. The strip showed purple colour in the presence of analyte cardiac troponin I. This presented method could be a powerful strategy for the quantitative detection of biomarkers with high

sensitivity (Bai et al., 2018). In an interesting work, a combination of acoustic and microfluidics was seen in the fabricated SERS detection platform. Surface acoustic waves were used to concentrate NPs at the periphery or the center of the glass capillary. This strategy improved the visibility of fluorescent outputs. ZnO nanorods and silver (as plasmonic NPs) were employed to achieve enhanced SERS signals. The developed method enlightens the future of point-of-care devices for distinct analytes (biomarkers) (Hao et al., 2020). Recently in 2020, a SERS-based detection platform was developed by metal NPs aggregated substrate. The polyacrylonitrile (PAN) nanofibers, thiol-modified purine (L)-polymer, and gold NPs are decorated as bio-essential ligands to develop a functional SERS platform. The developed sensing platform was used to detect uric acid concentration in body fluids. The method is highly sensitive and cost-effective and showed good stability (Saravanan et al., 2020). Furthermore, chronic kidney disease related biomarkers such as urea and creatinine were also targeted through SERS-based biosensor (Kukkar et al., 2023). For the successful detection of important biomarkers of dopamine, a digital SERS sensing platform was developed that facilitates the on/off mechanism for SERS signal analysis. The platform was coated first with 3D nanolaminate plasmonic crystals coupled with Raman-conjugated gold NPs. These nanostructured materials induce the additional hotspot that enhanced the SERS signals through molecular recognition of dopamine. This strategy can provide ultrasensitive results based on the SERS on/off mechanism with a detection limit of 1 pM (Nam et al., 2021). Interestingly, a self-assembly strategy using a thiol group and an on/off strategy in the heat map was employed to ensure the stability of the SERS immunoassay platform for adiponectin detection, a diabetes biomarker. The platform demonstrated a wide assay range, good reliability, a femto-scale limit of detection, and excellent selectivity without interference from other biomarkers. The reported biosensor showed potential of the developed technology for effectively detecting adiponectin levels in the biofluids of pregnant women. It was proposed as a clinical biosensor capable of diagnosing various obstetric diseases, particularly gestational diabetes mellitus, during early pregnancy (Kim et al., 2022).

The well-known pandemic also acknowledged as the global emergency posed by COVID-19, was showing the importance of identifying individuals with current or past infections as quickly as possible to prevent the further spread of the disease. In this context, SERS was described as an exciting technique for virus detection due to its ability to provide a fingerprint for any material. It has been used successfully with various COVID-19 virus subtypes, including Delta Cron and Omicron (Mousavi et al., 2022).

4.2. SERS-based detection of different cancer biomarkers

Early diagnosis of cancer is crucial for improving patient survival rates. Selective and sensitive detection of biomarkers in serum samples is vital for the early detection and dynamic monitoring of cancer. Several types of cancer biomarkers were also investigated through the SERS technique and developed various sensing platforms for their detection using these cancer-specific biomarkers. Among various types of cancer, prostate cancer is prevalent worldwide. Therefore, the detection of prostate-specific antigens, carcinoembryonic antigens, and alpha-fetoprotein enlightened the path to early diagnosis of cancer. These three biomarkers were simultaneously detected in a single test spot through the SERS-based vertical flow assay. Raman dyes encoded core-shell SERS nanotags were employed instead of conventional gold colloids. The developed method can be utilized as a point-of-care device due to its potential for rapid and highly sensitive SERS-based detection of multiple cancer-related biomarkers (Chen et al., 2019). The ratio of free prostate-specific antigen (f-PSA) to total prostate-specific antigen (t-PSA), represented as f-PSA%, is more accurate than t-PSA alone in diagnosing prostate cancer and distinguishing it from other prostate diseases. A study described a SERS-based biosensor to simultaneously detect t-PSA and f-PSA. The biosensor utilized 4-mercaptobenzoic acid

(MBA) on the immunocapture substrate, which was coupled to the t-PSA antibody through the carboxyl group. The combination of t-PSA induced Raman frequency shifts of MBA, enabling the detection of t-PSA. The biosensor employed immunocolloidal gold attached with f-PSA antibodies to selectively capture the f-PSA immobilized on the MBA-modified SERS substrates. The quantification of f-PSA was achieved through the SERS intensities of the 5, 5'-Dithiobis (succinimidyl-2-nitrobenzoate) (DSNB) probe. The results demonstrated good linear responses of f-PSA and t-PSA within specific concentration ranges. The biosensor combined Raman frequency shifts and intensities, simplifying the traditional procedures for f-PSA% detection. The developed biosensor has great potential for highly reproducible and accurate diagnosis of prostate cancer based on the results obtained (Zhao et al., 2023). Aptamer-based another strategy was also employed to develop SERS-based biosensor using a composite structure consisting of arrays of polystyrene colloidal sphere @Ag shell (PS@Ag) as a SERS-active chip. The chip was functionalized with a stepwise immobilization of a complementary DNA probe (SH-DNA) and a PSA aptamer (Apt). A Raman reporter called methylene blue (MB) was bound to the guanine base of the aptamer. The binding of PSA to the aptamer causes the release of the MB-Apt complex from the chip, leading to a decrease in SERS intensity of MB. The reduction in SERS intensity correlates with the concentration of PSA, enabling the quantification of PSA levels. The reported biosensor successfully detected PSA levels in serum samples, showing promise for clinical diagnosis, particularly in the context of prostate cancer (Zhao et al., 2022).

By combining the sensitivity of SERS-based biosensors with the power of intelligent vision, researchers potentially discovered digital biomarkers specific to hepatocellular carcinoma (HCC). These biomarkers can aid in early diagnosis, prognostic assessment, and monitoring of therapeutic response in HCC patients. It's important to note that while this approach shows promise, the development and validation of digital biomarkers for HCC detection using SERS-based biosensors and intelligent vision would require extensive research, rigorous testing, and validation on larger cohorts of patients. The discovered digital biomarker consists of 10 characteristic peaks in serum SERS spectra. The validation process involves unsupervised clustering of spectra from independent sample batches, comprising normal individuals and HCC cases. The results demonstrate high clustering accuracies of 95.71% for normal individuals and 100.00% for HCC cases. This non-biomarker-reliant molecular detection method is anticipated to facilitate mass cancer screening (Cheng et al., 2023). SERS-based biosensors have also been studied for breast cancer diagnosis by targeting a specific biomarker known as oncomiRs, which are specific microRNAs associated with cancer (Chauhan et al., 2022). Recently in 2023, a study combined the capture selectivity of molecularly imprinted polymers (MIPs) with the ultrasensitivity of SERS technique for the detection of colorectal cancer. This approach allowed the detection of nucleoside diphosphate kinase NM23-H2 or NDKB protein markers. The imprinted cavity of the MIP served as the only channel for the Raman reporter to approach the SERS substrate. The highly complementary non-covalent binding sites of the MIP selectively captured the target NDKB protein based on ionic, hydrogen bonding, or hydrophobic interactions. Specific recognition of the NDKB protein filled the imprinted cavity, hindering the Raman reporter's proximity to the SERS substrate, resulting in a significant decrease in the Raman signal. Other proteins of different sizes cannot match the imprinted cavity. The biosensor achieved ultrasensitive detection of NDKB through the change in the Raman signal. The LOD was reported to be 0.82 pg/mL, indicating its high sensitivity. The combined approach of MIPs and SERS offers several advantages, including low cost, fast response, high sensitivity, and selectivity (Lu et al., 2023). iREX (integrated Raman spectroscopic EXO) biosensor, which integrates SERS detection within a vertical flow diagnostic device introduced recently for the multiplexed quantitative profiling of exosomal proteins in clinical serum samples. Protein profiles of exosomes (EXOs) in clinical samples of cancer patients have emerged as promising

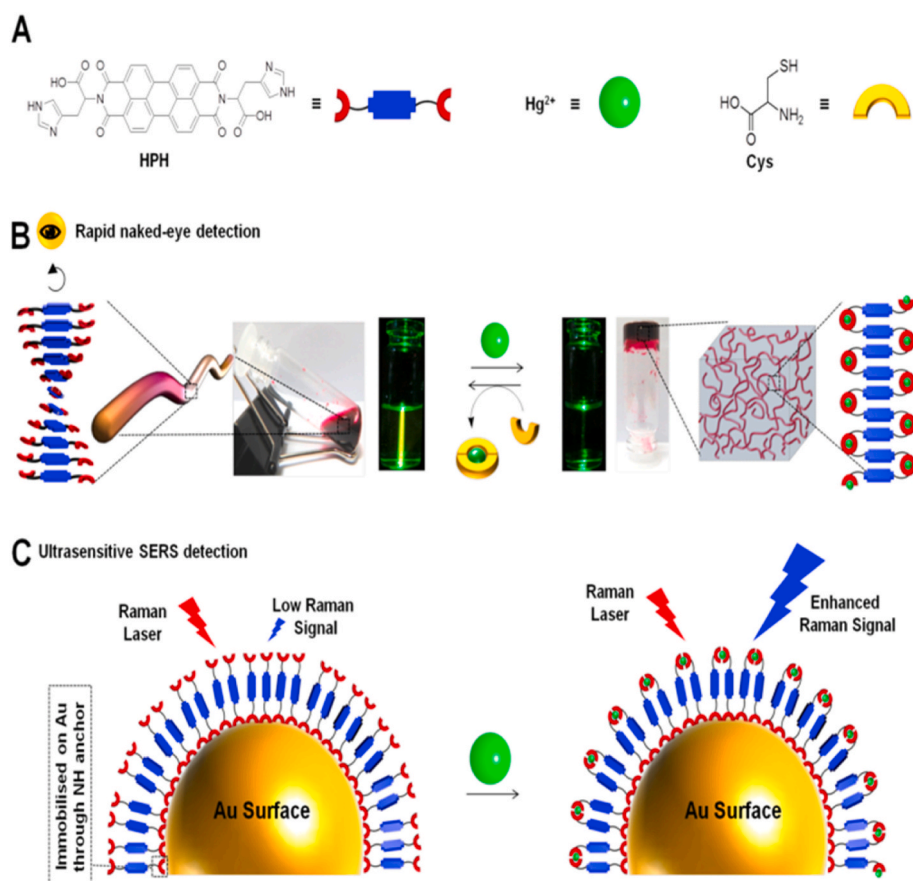


Fig. 2. Schematic representation of detection of mercury in water using HPH. (A) Molecular structures and corresponding schematic representations. (B) Rapid naked-eye detection of Hg²⁺; (C) Ultrasensitive SERS detection of Hg²⁺. Reprints with the permission of [Makam et al., 2018](#).

diagnostic and therapeutic biomarkers. However, the simultaneous quantitative analysis of multiple exosomal proteins of interest remains challenging. The iREX biosensor was used to quantitatively profile three exosomal proteins, MUC1, HER2, and CEA, in EXO samples derived from various breast cancer cell subtypes. The results revealed discriminative expression profiles of these proteins in different cell subtypes, enabling accurate diagnosis and molecular subtyping of breast cancer. The clinical utility of the iREX biosensor was further validated by quantitative analysis of MUC1, HER2, and CEA in patient blood serums. This aided in noninvasive breast cancer subtyping and longitudinal treatment monitoring. The iREX biosensor offered several advantages, including high sensitivity, molecular specificity, powerful multiplexing capability, and high diagnostic accuracy. It was believed to be a promising clinical tool for comprehensive analysis of exosomal proteins in clinical samples, contributing to personalized diagnosis and precise management of breast cancer ([Su et al., 2023](#)). Several recent studies have been targeted cancer related biomarkers such as miRNA ([Zhai et al., 2022](#); [Li et al., 2022](#); [Canning et al., 2023](#); [Weng et al., 2022](#)), Epidermal growth factor receptor, BRAF, programmed death-ligand 1 ([Dey et al., 2022](#)), squamous cell carcinoma antigen and cytokeratin 19 fragment antigen 21–1 ([Cao et al., 2023](#)) for the successful detection of different types of cancers using SERS technique.

4.3. SERS-based detection of organic dyes and drugs

SERS identified natural dyes and glazes present in art materials. Based on this, a SERS-based plasmonic sensor was reported showing an analyte enrichment system developed by direct laser writing technique. The sensing method was developed for the detection of water-dissolved analytes at trace concentrations. In this study, two types of systems were

combined to fabricate a device i.e., SERS active sensor site and analyte enrichment system. Both the systems were developed through the direct femtosecond laser printing method which is inexpensive and showed high performance. An analyte enrichment system was developed to support evaporating water droplets during the assay. The whole system was fabricated on a water-repellent surface as the substrate of the device namely polytetrafluoroethylene. This developed device was utilized to fingerprint various analytes based on SERS spectroscopy. Analytes like common organic dyes as well as medical drugs, are present in ultralow concentrations ([Pavliuk et al., 2019](#)). Another analyte i.e., kanamycin was also detected through the SERS-based aptasensor in the milk sample. The sensing platform was decorated with bimetallic gold and silver NPs which facilitates the binding of double-stranded DNA. Kanamycin-specific aptamers are immobilized on this NPs-coated surface. It was observed that with the increase in kanamycin concentration, the Raman intensity decreased significantly. Thus, the aptasensor showed a high sensitivity for the detection of kanamycin ([Jiang et al., 2019](#)).

Along with gold NPs, titanium dioxide was also employed to check its performance as a SERS signal enhancer. Two organic pollutants i.e., rhodamine 6G and methylene blue were detected by preparing a glass substrate coated with gold and titanium dioxide thin film that showed $\sim 10^7$ Raman intensity enhancement factors for detection of rhodamine 6G and methylene blue. They are capable of inducing photodegradation efficiency for methylene blue under sunlight in approximately 40 min. Thus, these nanomaterials were found to improve optical absorption and high sensitivity toward SERS-based detection of organic pollutants ([Singh et al., 2019](#)). To determine the concentration of rhodamine 6G in a mixture solution, a paper-based sensing platform coated with silver NPs on polymer-grafted cellulose was fabricated with the function of

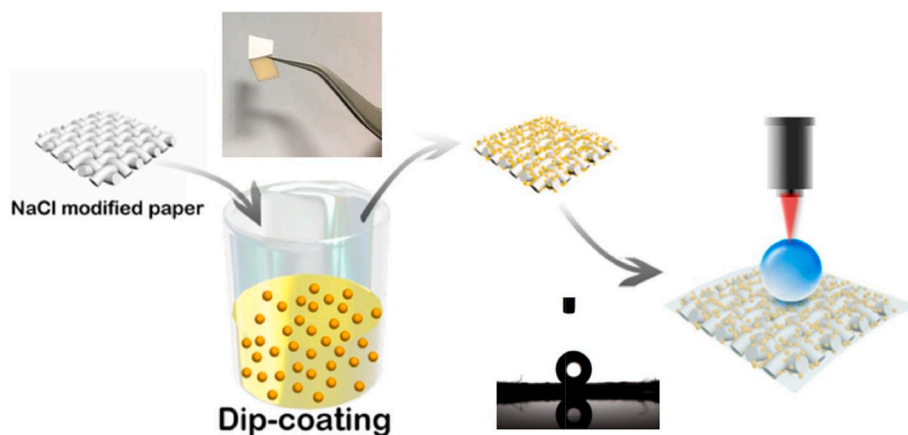


Fig. 3. Schematic illustration of the fabrication and analysis process for hydrophobic paper-based SERS platform. Reprints with the permission of Zhang et al., 2019.

analyte enrichment. It was reported that silver NPs with 30 nm size at 532 nm excitation showed excellent Raman signals (Trang et al., 2021). Another study employed molecularly imprinted polymer (MIP) as a biorecognition element in SERS-based detection platform which is very cost-effective, simple, and sensitive. Researchers assembled MIP thin film with gold NPs for SERS spectroscopic detection of polycyclic aromatic hydrocarbons. When the interaction of polycyclic aromatic hydrocarbons with specific MIP takes place close to gold coated surface, the Raman signal of a molecule is enhanced due to plasmonic NPs. The performance of the developed sensing platform was investigated in creek water and seawater samples to detect pyrene. This type of sensing platform can be utilized for food safety, environmental control, and biomedicines (Castro-Grijalba et al., 2020). The gold NPs are the first choice of researchers to be used in the SERS-based detection of analytes. On the contrary, some researchers utilized silver and reduced graphene oxide (rGO) also for the successful detection of multiple analytes simultaneously. A silver and rGO decorated sensing platform were fabricated recently for the detection of multiple analytes such as 4-mercapto benzoic acid, Rhodamine-6G, and 2,4-dinitrotoluene. The results from this study showed good homogeneity, enhanced Raman intensity, nanomolar sensitivity, and stability. Thus, the study showed that silver and rGO can also be a good choice to be employed in SERS-based investigation of biomolecules (Naqvi et al., 2019). An anti-epileptic drug named Perampanel was quantitatively detected through SERS-based sensor. In this study, the pulsed laser deposition method was used for the synthesis of roughened gold NPs films. The linear range was calculated as $3 \times 10^{-7} \text{ M} - 3 \times 10^{-6} \text{ M}$. From this study it was elucidated that the SERS intensity is dependent upon the analyte concentration as well as the Au surface and solution contact time (Zanchi et al., 2020).

4.4. SERS-based detection of inorganic ions

Makam. P et al. adopted a SERS-fluorescence-based sensitive method for the detection of mercury in drinking water samples. They fabricated a SERS-based platform coated with gold and polystyrene's monodispersed nanosphere monolayer which improves the selectivity and very low detection limit for sensing mercury in water [Fig. 2]. Simple histidine conjugated perylene diimide bolaamphiphile was employed as a dual optical marker in this study for visualization of fluorescence in the SERS-based sensing platform. These optical markers are described as highly sensitive and selective. The limit of detection was achieved as 60 aM (aM) which is quite low from existing strategies developed for mercury detection. Therefore, this method can be used for ultrasensitive detection of other toxins as well, present in drinking water (Makam et al., 2018). Mercury ions in drinking water are also detected through another SERS-based sensing platform developed through the coating of silver NPs functionalized with 4,4'-dipyridyl. They utilized capillary

action force to collect samples and a portable Raman spectrometer for onsite analysis. Although it can be used as a real-time analysis device, but it showed high limit of detection (0.1 ppb) as compared to sensor developed by Makam and team (Zhao et al., 2020).

4.5. SERS-based detection of toxins

SERS-based sensors are applicable with various types of biorecognition elements such as aptamers (aptasensor), DNA (DNA sensor), antigen-antibodies (immunosensor), molecularly imprinted polymers (MIP-based sensor), etc. In this context, a study performed with aptamers for sensing microcystin-LR spiked in a sample of tap water. Aptamers are highly specific short oligonucleotide fragments that are employed in SERS-based sensing strategy. The substrate was fabricated by utilization of microcystin-LR complementary DNA conjugated with gold and magnetic NPs. Aptamer combined with gold NPs and cDNA conjugated with magnetic NPs were employed as signal probes and capture probes in this study. Although aptamer synthesis is quite a tedious task but aptamer-based detection is highly reliable and sensitive and thus used widely. The detection limit was obtained from this method as 0.002 ng/mL with a linear range of 0.01–200 ng/mL (He et al., 2019). Toxins like Ochratoxin-A (OTA) and aflatoxin-B1 require rapid monitoring due to their severe health effects. The SERS-based sensor was fabricated recently for the detection of these two toxins. The SERS enhancement factor was optimized by adjusting the pH used to synthesize silver NPs which was obtained at pH 11. The limits of detection resulted as 2.63 pg/mL and 4.15 pg/mL for Ochratoxin-A (OTA) and aflatoxin-B1 respectively (Kutsanedzie et al., 2020). A hydrophobic surface was prepared to develop gold NPs decorated SERS-based detection platform. The study showed a double-decker NPs-coated structure arranged on the wax-coated hydrophobic surface which served as an inducer of aggregation as well as the concentration of analyte-mixed gold NPs in the aqueous phase. This is a very simple yet sensitive method to detect hazardous chemicals present in food and the environment (Wu et al., 2018).

Furthermore, the SERS method also exploited to detect chemical warfare agents as well as toxic industrial chemicals (Lafuente et al., 2021). The explosive chemicals namely, dinitrobenzenethiol and half mustard agents have been detected through this method successfully (Stuart et al., 2006).

4.6. SERS-based detection of pesticides

The presence of pesticides in food-related items is a grave concern in recent years. Various strategies have been developed for the detection of their traces in food as well as in the environment (Jain et al., 2022; Singh et al., 2020; Pundir and Chauhan, 2012). Distinct types of NPs play an

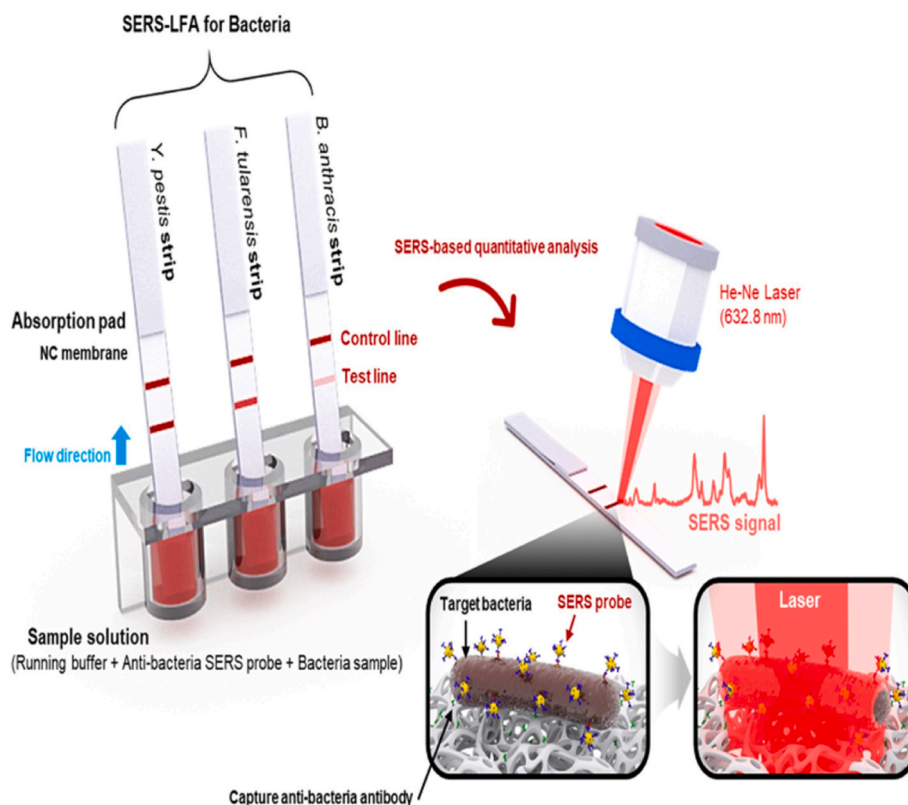


Fig. 4. Operating principle of the SERS-based LFA sensing platform. LFA strips for *Y. pestis*, *F. tularensis*, and *B. anthracis* were dipped into wells of a 96-well ELISA plate containing mixtures of SERS nanotags and different concentrations of bacteria in buffer solution. The immunocomplexes that formed migrated by capillary action toward to the test line, where their Raman signals were measured and analyzed. Reprints with the permission of Wang et al., 2018.

important role in SERS-based detection. Among them, gold is extensively employed in most of the studies. Orderly spaced gold NPs supported by mesoporous silica are utilized as substrates for the development of SERS-based sensors to detect pesticides in food samples. Pesticides such as thiamethoxam, 2,4-dichloro phenoxy acetic acid, and pymetrozine were detected by SERS spectra with excellent sensitivity, and stability. The synthesis of the substrate includes in-situ reduction, ion exchange, and self-assembly techniques to arrange gold NPs inside the silica channels (Xu et al., 2020a). This technique seems simple and reliable for the detection of pesticides in food samples. The validity of the developed method was also confirmed by HPLC satisfactorily. rGO and gold nanostar nanocomposite-based sensor was fabricated to detect acetamiprid traces in green tea samples through the SERS technique. SERS active nanocomposite decorated sensing surface was combined with different concentrations of acetamiprid pesticide in a linear range of 1.0×10^{-4} to $1.0 \times 10^3 \mu\text{g/mL}$. The sensor showed good accuracy and precision high stability and can detect multiple pesticides at a time (Hassan et al., 2019). Another advanced SERS strategy has been utilized for the successful detection of two common herbicides namely carben-dazim and thiabendazole. They used leaf-inspired biomimetic material by soft lithography method to assemble gold NPs for creating electromagnetic hotspots. These biomimetic surfaces influenced the performance of SERS (Sharma and Krishnan, 2018). There are various types of SERS-based sensors developed recently for the detection of different types of pesticides like organophosphorus (Yaseen et al., 2019), thiram etc. A recent study introduced a novel hydrophobic-SERS substrate platform based on silver nanoparticles (AgNPs) functionalized commercially available filter paper [Fig. 3]. The platform offered several advantages over conventional substrates, including simple and large-scale preparation, direct droplet-detection, and reusability. The study investigated the fabrication, physical characterization, and SERS sensitivity of the substrates for food determination. The experimental

results demonstrated high reproducibility, with less than 9% spot-to-spot variation in Raman intensity. Additionally, the platform was successfully applied to detect melamine in diluted milk, exhibiting an instrument detection limit (IDL) of 1 ppm in a linear correlation ranging from 1 ppm to 1000 ppm. (Zhang et al., 2019). The MIP-based sensor attracted most researchers because it can replace costly antibodies and enzymes (Ma et al., 2021). Another MIP-based SERS sensor was fabricated in 2020 for the detection of pesticides namely 2, 4-D, in water and food samples. Magnetic-based MIP NPs along with SERS active gold NPs were decorated to develop a sensing platform. They successfully achieved the limit of detection as 0.00147 ng/mL. The results were also compared with the conventional, yet highly sensitive technique HPLC and the developed sensor showed high sensitivity as compared to HPLC (Xu et al., 2020b).

4.7. SERS-based detection of bacteria

The same combination of SERS and lateral flow assay strips as mentioned earlier was used again for the detection of bacterial pathogens. To increase the sensitivity of SERS, researchers exploit SERS nanotags namely “Raman reporter-labeled gold NPs” that are an alternative to conventional gold NPs. These reporters provide information about bacterial presence by changing the colour of the strip’s test line [Fig. 4]. Furthermore, the Raman peak intensity was observed through coated SERS nanotags. The described method is sensitive and took less time for the detection of bacteria (Wang et al., 2018).

SERS has also exploited for the successful detection of harmful bacteria through several studies. To detect *E. coli* O157:H7 in the water sample, gold NPs coated starch magnetic beads functionalized with specific antibodies (SERS hotspot), were prepared recently. Antibodies were immobilized on the sensing surface by a linker protein, “4× gold-binding peptide-tagged Streptococcal protein G” that also performed the

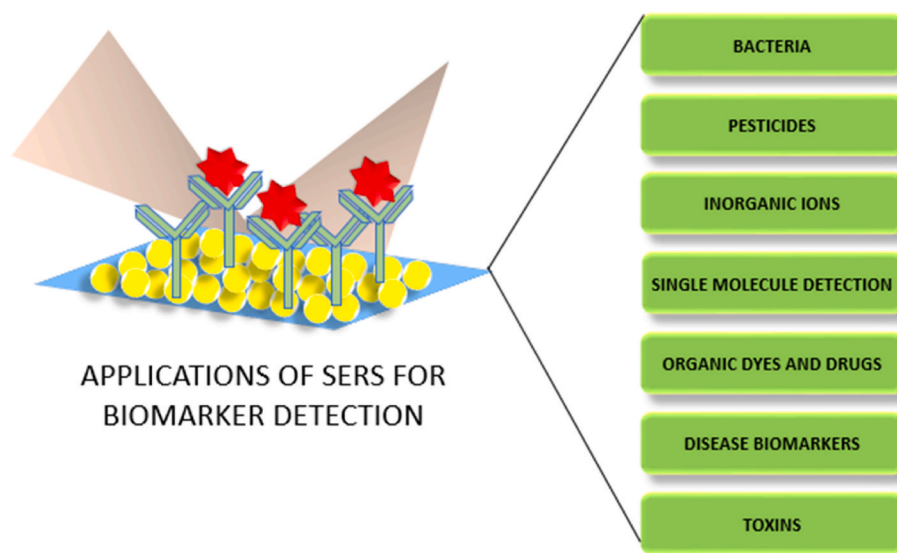


Fig. 5. A pictorial representation of SERS applications for biomarkers detection.

Raman reporter function by exhibiting fingerprint signals. The concentration range detected was reported as 10^0 – 10^5 CFU/mL with a low limit of detection. This type of sensitive device is important to study pathogenic bacteria in food and water samples (You et al., 2020).

4.8. SERS-based detection of single-molecule detection

The single molecules that are assembled to form molecular machines are essential for a variety of biological activities. As a result, information regarding various diseases caused by their imbalance and internal mechanisms of cell activity can be obtained by their quantification (Chauhan et al., 2022). The SERS technique is extensively used for single molecule detection for 15 years with a very low limit of detection. SERS also enables single molecule detection very rapidly with high specificity. Silver and Raman labeling compound-coated gold NPs combined with silicon dioxide were utilized as SERS active materials. Enzymes activate the reduction of silver ions and growth on SERS active nanomaterials which ultimately increases SERS signals. Ascorbic acid was responsible for forming a silver hot spot on the nanomaterials sensing surface in the detection range of 0.6–9.0 ng/mL and the very low limit of detection (0.09 ng/mL) (Pham et al., 2020). Another study highlighted the single-molecule nucleobase sensing based on the SERS spectroscopy technique. SERS in combination with nanofluidic facilitated the electrokinetic capture of nucleobase and local identification through direct Raman spectroscopic fingerprinting. Plasmonic nanoslit was used in this study which was optically engineered and locally enable SERS (Chen et al., 2018). SERS in combination with other techniques shows very significant results for the sensing of certain chemicals, for example microfluidic-SERS used to overcome surface interaction challenges. SERS is also used in conjunction with electrochemistry to reveal the behavior of molecules in various states of oxidation. The surface of SERS and electrochemical electrodes are basically made up of metals that allow the spectroscopic detection of electrochemically active molecules (Hakonen et al., 2015). The advantages of single-molecule SERS include decreased sample bleaching and fingerprint-like information of chemicals. The single molecule SERS technique is usually employed in fundamental research for the detection of resonant dyes. Although the technique was discovered decades ago, its applications outside the labs are quite recent in various fields. Fields like art preservation utilized SERS as it can detect very low levels of chemical concentrations. Fig. 5 shows a summary of SERS applications in detection of different biomarkers.

5. Conclusion

SERS is a highly sensitive and attractive technique for the detection of a variety of analytes and provides structural information of any molecules present even in traces. SERS-based sensing platforms have been successfully exploited in the field of chemistry, physics, biology, nanoscience, and medicines for interdisciplinary applications. We have reviewed recent studies demonstrating the versatility of SERS in the detection of analytes. Several studies have been carried out to develop SERS-based sensors for distinct analytes like biomarkers, single molecules, pesticides, and so on. It has been seen that SERS-based biosensors exhibit very low detection limits along with high sensitivity as compared to others. SERS in combination with other advanced techniques like laser printing, microfluidics, etc. can improve the performance of sensors several times. Common SERS incorporated with some resonance enhancement techniques like an optical waveguide, optical coherence, and resonant adsorption showed enhanced capabilities for ultrasensitive detection of analytes. However, its use is restricted in real-time analysis due to some challenges like limited output wavelength and difficulties in matching analytes vibration (electronic transition frequency) with plasmonic frequency. Furthermore, the variety of biorecognition elements like MIP, and aptamers are also giving satisfactory results in the fabrication of SERS-based biosensors. At its most complex level, distinct types of single molecules can be easily and routinely observed by SERS-based sensors due to the large enhancement. Additionally, SERS is an exceptional technique to characterize small numbers of analytes presented to or near plasmonic surfaces.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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